

1.1 Learning from Nature

1. According to the textbook, “nature” is best defined as

(a) All structures designed and built by humans (b) The physical, natural and material world of life that exists without human intervention (c) Only the weather and climate of a given area (d) Only the living organisms such as plants and animals

3. Organisms have well-adapted structures that allow them to ——— on their immediate environment

(a) build (b) depend (c) interfere (d) survive

5. Using imitation of nature, engineers can develop more efficient technologies at ——— scales

(a) only macro (large) (b) macro (large) and nano (very small) (c) only micro and milli (d) only theoretical

7. Nature provides us with doable structures and forms which can be developed as

(a) functional and applicable mechanisms (b) decorative items only (c) genetic copies of living organisms (d) artificial habitats only

1.1 Technologies Imitated from Nature

8. Weaving technology was developed by studying the formation of **webs** of the

(a) termite (b) swallow (c) spider (d) mosquito

10. The East Gate Center building in Harare was designed by imitating the

(a) flight of birds (b) beak of a kingfisher (c) mound architecture of termites (d) shell of shellfish

12. The first heavy aircraft technology was designed by imitating

(a) kangaroos (b) birds (c) elephants (d) mosquitoes

2. Which of the following is part of nature as described in the textbook?

(a) A photograph camera (b) Landscape sceneries (c) A bullet train (d) An injection needle

4. Scientists and engineers learn from nature mainly through ——— of the physical structures, shapes, materials and functional mechanisms of natural facts

(a) imitation (b) elimination (c) classification (d) translation

6. In the textbook, nature is described as a reliable source of

(a) water and electricity (b) money and labour (c) only human-built tools (d) knowledge, ideas and concepts

9. Swallow nests have inspired the

(a) design of aircraft wings (b) design of injection needles (c) design of house building (d) making of bulletproof vests

11. Termite mounds that inspired building designs have

(a) chimneys with constant temperature and humidity (b) sharp teeth for cutting (c) webs for catching insects (d) trunk-like parts for grasping

13. Aircraft engineers designed aircraft wings and their flight techniques by imitating the wings of

(a) mosquitoes and spiders (b) birds and bats (c) bees and ants (d) swallows and termites

14. The technology of the photograph camera was made by engineers imitating the

(a) elephant trunk (b) human eye (c) human ear (d) mosquito proboscis

16. Injection needle technology was copied from the ——— of a mosquito

(a) proboscis (b) wing (c) eye (d) leg

18. A cutting saw was imitated from the nature of the sharp ——— of animals

(a) claws (b) horns (c) beaks (d) teeth

20. Ceramics are an imitation of the nature of crack-resistant

(a) shellfish exoskeleton (b) spider web (c) termite mound (d) elephant trunk

15. The movement of bio-robots was designed from the movement of a

(a) kingfisher (b) swallow (c) kangaroo (d) termite

17. The sensors of electrical devices were imitations of biological

(a) bones (b) muscles (c) neurons (d) arteries

19. Synthetic bulletproof vests are imitations of the silk of

(a) mosquitoes (b) spiders (c) termites (d) kangaroos

21. A robotic arm was imitated from the ——— of an elephant

(a) tusk (b) ear (c) trunk (d) foot

1.2 Biology, Technology and Biotechnology

22. Biology is the study of the structures, functions, growth, origins, evolution and ——— of living organisms

(a) distribution (b) destruction (c) decoration (d) disposal

24. The blend of biology and technology forms

(a) biochemistry (b) biotechnology (c) biophysics (d) bioinformatics

26. Which of the following is given in the textbook as an example of biotechnology?

(a) Using a thermometer to measure fever (b) Using a GPS to collect data (c) Using yeasts to produce bread and beer (d) Using a microscope to magnify cells

23. Technology is the application of scientific knowledge, skills, methods and processes for the ——— of devices and tools used in scientific investigations

(a) sale (b) repair (c) description (d) production

25. Biotechnology is a technology that utilizes ——— to develop useful products

(a) radioactive tracers (b) mechanical gears (c) electrical circuits (d) biological systems

27. Biotechnology is described as the integration of natural and ——— sciences

(a) economic (b) engineering (c) language (d) historical

1.2.1 Benefits of Biology to Technology

28. According to the textbook, biology is a source of materials that enable us to imitate, design, adapt and ——— modern technologies

(a) reject (b) destroy (c) quarantine (d) develop

30. Gene engineering is used to

(a) build chimneys (b) cure genetic conditions (c) weave spider webs (d) make cameras

29. Many biochemical substances in plants have been used to develop

(a) medicines (b) bullet trains (c) robots (d) cutting saws

31. Biosynthetic materials are created mainly to

(a) repair damaged body tissues (b) design termite mounds (c) form the wings of aircraft (d) measure body temperature

32. The noses (fronts) of bullet trains were designed by imitating the shape of the beak of the ——— when it dives into water

- (a) swallow (b) crow (c) eagle (d) kingfisher

1.2.2 Uses of Technology in Biology

33. The uses of technology in biology are the ways in which technological tools can be applied to

- (a) replace biological systems (b) end scientific investigations (c) solve various human problems (d) reduce human services

35. Studies that provide detailed information on the chemical components of medicinal plants are called

- (a) biophysical studies (b) environmental studies (c) bioinformatics (d) biomedical studies

37. Bioinformatics is a scientific discipline involving computer technology to collect, store, analyze and ——— biological data and information

- (a) delete (b) dilute (c) damage (d) disseminate

39. A digital thermometer is an instrument used for measuring

- (a) body temperature (b) blood sugar (c) body fat (d) blood pressure

41. In a pregnancy urine test, the appearance of two lines on the test shows that

- (a) the woman is not pregnant (b) there is a bacterial infection (c) the test strip is broken (d) the woman is pregnant

43. An HIV test is a detector used to

- (a) measure body fat (b) magnify objects (c) identify whether a person is infected with the virus (d) map landscapes

45. The device used for investigating information on diseases or cancerous areas of the human body is

- (a) GPS (b) a digital thermometer (c) Computer Information Technology Scanning (CITS) (d) a handheld body fat calculator

47. A Positron Emission Tomography (PET) scan uses a special dye containing ——— that is swallowed, inhaled or injected into an arm vein

34. Biochemical studies are helpful to investigate information on carbohydrates, proteins, lipids and

- (a) body fat (b) blood sugar (c) water (d) nucleic acids

36. The science of using physical devices to gather biological information at all scales of biological organization is called

- (a) biophysical studies (b) biochemical studies (c) biomedical studies (d) biogeographical studies

38. The study of the distribution of species along with geographic ecosystems through geological periods is called

- (a) bioinformatics (b) biomedical studies (c) biophysical studies (d) biogeographical information

40. A pregnancy urine test detects the amount of the hormone ——— produced in the placenta

- (a) insulin (b) Human Chorionic Gonadotropin (HCG) (c) adrenaline (d) melatonin

42. The diabetic blood test measures the level of ——— by taking a sample of blood from the finger tip

- (a) blood sugar (b) cholesterol (c) body temperature (d) oxygen

44. Microscopes are used to

- (a) measure body temperature (b) magnify objects (c) detect pregnancy (d) collect geographical information

46. Computed Tomography (CT) scan is a sophisticated ——— technology used to take many X-ray pictures of the body

- (a) X-ray (b) laser (c) ultrasound (d) infrared

48. GPS (Geographic Positioning System) is a device used to

(a) antibodies (b) vitamins (c) enzymes (d) radioactive tracers

(a) measure the level of blood sugar (b) collect biogeographical information on landscape, plants, animals and human movements (c) detect pregnancy hormones (d) treat bacterial infections

1.3 Impacts of Biology and Technology on Society and Nature

49. In the textbook, the impacts of biology and technology refer to factors that pose ——— on the society and the natural world

(a) only positive effects (b) only negative effects (c) positive or negative effects (d) no effects at all

51. According to the textbook, antibiotics were created to treat

(a) viral infections only (b) plant diseases (c) insect bites (d) bacterial infections

53. According to the textbook, technology has damaged our world in two main ways through

(a) biotechnology and bioinformatics (b) medicine and education (c) pollution and the depletion of natural resources (d) agriculture and transportation

1.4 Ethical Issues in Biology

55. Ethical issues in biology arise concerning the rightness or wrongness of using ——— for the health and wellbeing of humans

(a) biological discoveries (b) technological tools only (c) X-ray machines (d) fertilizers

57. Which of the following is an ethical treatment of plants according to the textbook?

(a) Cutting tree terminals for experiments (b) Caring for plants and planting trees (c) Testing seeds with destructive rays (d) Burning forests for farmlands

59. The three Rs principles for the ethical treatment of research animals are

(a) Reduction, Recycling and Reuse (b) Reduction, Refinement and Replacement (c) Research, Respect and Rescue (d) Reproduction, Rest and Reaction

50. Which of the following is an impact of biology on society given in the textbook?

(a) Polluting clean air with industrial emissions (b) Depleting natural resources (c) Ensuring food security through an increase in productivity (d) Causing acid rain

52. Which of the following is given in the textbook as a negative impact of biology on the natural world?

(a) Biological weapon production and use (b) Improving health and longevity (c) Ensuring food security (d) Achieving a better supply of energy and clean water

54. According to the textbook, biochemical agents, pesticides and fertilizers cause

(a) a better supply of energy (b) toxicity and biohazards (c) improved food security (d) constant humidity in buildings

56. According to the textbook, trees properly function in ——— forms without any centralized order-providing unit (brain)

(a) decentralized (b) centralized (c) artificial (d) computerized

58. Animals are described as ——— creatures that have a nervous system, feel painful feelings, and have interests and values

(a) sentient (b) mechanical (c) insensitive (d) artificial

60. Which of the following correctly describes the ethical principle of Replacement?

(a) Improving experimental techniques to reduce the number of animals used (b) Substituting experiments on animals with alternative techniques such as cell cultures and computer simulated models (c) Reducing the suffering of

animals using less invasive techniques (d) Using more animals in experiments to get better results